

README



How to Join a Basler blaze camera and a Basler ace/ace 2 Camera

Applicable to blaze, ace and ace 2 camera models only

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1 Introduction

This readme describes how to build a joint camera system consisting of a Basler ace/ace 2 area scan camera and a Basler blaze 3D ToF camera. Such a camera assembly will allow you to create color point clouds or add high-resolution 2D images to your 3D application to enhance your pattern recognition or object detection algorithms.

The readme lists the components required as well as installation instructions.

The assembly includes a mounting bracket for joining the two cameras that has been designed by Basler specifically for that purpose. In addition to the **.stp** and **.stl** files included in the package, this readme also contains some recommendations for 3D printing.

2 Hardware Requirements

Description	Quantity	Notes
Camera, Basler blaze-101	1	Order number 107796 (Rev. 08+)
Camera, ace GigE or ace 2 GigE	1	(see recommendations below)
Lens, Basler C125 series 4/6/8 mm	1	(see recommendations below)
Socket screw, M3 x 14, ISO 4762	3	
Socket screw, M4 x 35, ISO 4762	2	
blaze & ace mounting bracket	1	See Suggested Printing Parameters



The design of this mounting bracket is intended for test and evaluation purposes only. It's not recommended for mass production.

2.1 Camera and Lens Recommendations

The mounting bracket is suitable for any ace or ace 2 GigE camera. However, Basler recommends the following components for most RGB-D applications:

- acA1300-75gc ace camera
Order number: 106757
- acA1920-51gcBAS ace 2 camera
Order number: 107823
- C125-0418-5M (f 4 mm, F 1.8) lens
Order number: 2000034830

Alternatively, any other ace or ace 2 GigE camera can be used. Since the mounting bracket was designed specifically for use with Basler C125 lenses, Basler suggests using a lens from that series for best mechanical fit.

Choose any of the following lenses according to the requirements of your application:

- C125-0418-5M-P (f 4 mm, F 1.8); order number 2000034830
- C125-0618-5M-P (f 6 mm, F 1.8); order number 2000034831
- C125-0818-5M-P (f 8 mm, F 1.8); order number 2000034832

If you choose a different lens, make sure its mechanical dimensions fit the mounting bracket.

2.2 Suggested Printing Parameters

The mounting bracket has been designed and tested with the following printing parameters:

- 3D printer type: FDM (fused deposition modelling)
- Filament material: PETG, PLA
- Supports: Yes (on print bed)
- Rafts: No
- Z-resolution: 200 µm
- Infill: 40 % (gyroid)
- Position the mounting bracket model on the print bed with the side that will be facing the blaze camera pointing down.

3 Assembly Instructions



1. Attach the lens to the ace/ace 2 camera.
2. Set the aperture ring to the desired aperture.
3. Adjust the focus.
4. Lock the focus ring and the aperture ring in place using grub screws.
Alternatively, you can also use adhesive.
5. Remove the thumbscrews from the focus and aperture rings and leave only the grub screws mounted.
6. Attach the ace/ace 2 camera to the mounting bracket using the M3 screws.
7. Attach the mounting bracket to the blaze camera using the M4 screws.



The mounting bracket has been designed for minimum distance between the optical axes of both cameras. Therefore, the thumbscrews of the Basler C125 lens have to be removed as they would interfere with the body of the blaze camera or the mounting bracket. Instead, use the grub screws provided with the C125 lens to fix the focus and aperture rings.

Revision History

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AW00166601000	1 Feb 2021	Initial release version of this document.